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## MANAGING CONSUMERS' HEALTH IN THE HEAT AND SMOG

Introduction: As the temperature rises, and the smog level increases, there is the potential for serious health concerns that staff must be aware of and address. This inservice will attempt to cover some of these concerns and point out antecedent and follow-up actions that should be taken to maintain the health and safety of our consumers.

### General Concerns:

As the temperature rises, people's comfort-level becomes challenged. The smog level increases with the heat and this adds to the discomfort. Staff attention must be paid to providing adequate cool water (not soda) intake, checking for weather appropriate clothing, protecting skin from the sun, not overdoing the activity level, and assessing the comfort needs of each consumer.

- Providing adequate fluid intake involves providing sufficient opportunities for the consumers to drink fluids. Staff should provide two pints of cool water per consumer per day, more if necessary. (A typical bottle of drinking water is a pint.) Water can be purchased with HDS program money and stored in a cooler with ice or freezer packs in the Job Coach's car and carried in bottles in the backpack, and drinking fountains can be used. Additionally, consumers should be encouraged to eat breakfast and drink fluids first thing in the morning. The Job Coach should provide opportunities to drink additional fluids throughout the day.

- Checking for whether appropriate clothing involves inspecting the dress of the consumers as they join the program and making corrections before leaving their home. People should wear clothing that is light in color and weight, allow for ventilation, and cover the skin to prevent sunburn. Hats should be encouraged and sunglasses allowed.

- Protecting the skin from the sun involves adequate covering and the use of sunscreen. Clothing/hats are the simplest protection. Sunscreen may be purchased with program money and should bear a code number higher than 15. It should be applied before the consumers go into the sun.

- Activity choices should be directed toward shady outdoor settings, and indoor ones (particularly with air conditioning). Heavy work-related activities should be scheduled first thing in the morning before it gets hot. Work alternatives can be politely asked for that involve less heavy work or indoor work. An activity involving using an air-conditioned setting should be planned for daily. This can include the library, bowling alleys, stores and malls, movie theaters, restaurants, etc.

- Assessing the comfort needs of each consumer involves asking about and observing them for comfort related issues. If a person appears affected by the weather,

N they should be given less demands, opportunities to rest, extra fluids, or whatever appears reasonable. They should be touched to determine if they are hot or feverish and observed for lethargy or sleepiness. They may require medical evaluation and the HDS office should be notified (see below). Medical Concerns: People challenged with seizures, psychiatric issues, high blood pressure and other diagnoses can react to the heat and smog much more severely than other persons. Staff must be aware of the medical issues associated with their consumers and of the specific issues the heat and smog may raise in their care.

- Many medications cause "photosensitivity" which involves making the person taking the medication susceptible to sunburn and rash with even moderate exposure to the sun. Some medications taken by many of our consumers with this side effect include: Tegretol, Seroquel, and Risperdal. Staff will look for this information in the "side effects" section of each HDS report. Management of this problem includes covering the skin with clothing/hats and use of sunscreen.

- Persons diagnosed with seizures have a higher likelihood for experiencing seizures as the heat/smog level rises. They should not be overheated by exertion or overexposure to the sun. They should be allowed to sit out work, spend time in air-conditioned settings and watch for signs of seizure activity. Any report from the consumer, or direct observation, that they are not feeling/acting normal should be responded to with a call to the HDS office. They may be sent home or, in extreme circumstances, be taken by ambulance to an emergency room.
- Persons taking the medicine Lithium Carbonate are taking salt and should be encouraged to consume adequate fluids. Failure to do so can result in toxic symptoms. These include diarrhea, vomiting, tremor, mild ataxia, drowsiness, or muscular weakness. Giddiness, ataxia, blurred vision, tinnitus (ringing in ears) and a large output of dilute urine are signs of higher-level toxicity. A physician should be notified if any of these symptoms are observed.
- Persons with excessive perspiration or no perspiration at all, a fever, feeling weak, acting sleepy, etc. may be reacting to the heat/smog. Any deviation from their acting their "usual selves" should be looked for. The HDS office should be called by the Job Coach to discuss the observed symptoms. The consumer may be given light duty, taken home, or taken to an emergency room depending upon the conditions. Please consider that we will not be criticized for acting too conservatively in protecting the health of our consumers. If you suspect a problem, do what you need to solve it before it becomes worse.

## A Near Tragedy

In the summer of 2005, one of our groups in Pasadena had a consumer who we have served for many years that became so severely affected by the heat that he nearly died. Even after our efforts to provide water and fluids, to seek low demand activities and shade, to use an air-conditioned car and keep a good watch over him, he went unconscious within blocks of his drop-off from apparent heat stroke. He was immediately taken by ambulance to a nearby hospital after being placed in the Intensive Care Unit for several days until he made a full recovery. There were lessons learned from this near tragedy that need to be considered by all HDS caregivers.

- This consumer was on a lot of medications that caused him to be sedated, sleepy, have poor balance (ataxia) and appear "droopy" even on cool days. This medication interfered with his ability to perspire normally, to be alert enough to drink and eat normally and masked the symptoms of heat related problems. Staff must be extra careful with consumers who are taking a lot of medications and are very sedated on a regular basis. They are most likely to be affected by the heat. The HDS office must be made aware of this problem in order to look into the appropriateness of serving these individuals on very hot days.

- This group had someone else in it who was intolerant of many community settings with people in it and would holler loudly, tantrum and become aggressive in these settings. Consequently the group spent their days visiting various parks and open settings. On hot days they were sitting in the shade, but still in the heat and humidity. Efforts were needed to seek out air conditioning for some of the activities. Libraries, stores/malls, bowling alleys, restaurants, and an HDS office were all possibilities. Swimming pools are open as are gyms in large parks. Without the opportunity to really cool off (remember, their bedrooms often do not have air conditioning), the body's temperature can gradually rise to a fever and proceed to heat stroke over a period of days.

- The consumers must be strongly encouraged to drink water (not soda) throughout the day. The cooler the fluids are kept, the better. The consumers should be observed to see that they are perspiring normally, they should be touched to see if they are running a fever, they should be talked to often to see if they are alert and how they are feeling. If they are not perspiring, are hot to the touch, or are lethargic and "spacey" the HDS office must be notified. Staff may be instructed to cool them down with water, take them to the HDS office, to their home, or even call "911".

- Staff will routinely offer to "wet down" consumers when outside for activities. They may wet their hair, hat, shirts, arms and legs, etc., in an effort to promote keeping cool. Normal perspiration is impaired by medications people are taking so getting wet is the next best thing. HDS caregivers must make every effort to plan for the heat of the season when serving our consumers. Staff "common sense", initiative and prompt intervention can save lives. The HDS administration will do everything possible to facilitate this process.

## **POLICIES FOR SUPPORTING CONSUMERS DURING HIGHER TEMPERATURES**

As the maximum daily temperature exceeds 85-90 degrees (usually in May-June) the Executive Director and the Program Director will identify the at risk consumers for heat and smog issues. These will be consumers with high levels of medications that can affect the body's ability to deal with the heat, people with partially controlled seizures, and people who are morbidly obese, elderly or otherwise medically fragile. High levels of medications will include two or more Neuroleptics, mood elevators, blood pressure medications and/or seizure medications. These people are often photosensitive, falling asleep during part of the program day and may already have some restrictions on their participation in general activities. The identified consumer's activity schedule will be scrutinized and plans made for adequate opportunities to rest, drink adequate water, and visit settings with air conditioning for a minimum of three hours a day. Their direct care staff will receive supplemental training on supporting them during the heat, their group compositions may be adjusted to permit extra support and their health status monitored more closely. Additional advice from their physicians may be requested to determine how best to serve them during higher temperatures. During this period of higher temperatures, the overall day program expectations will shift to ensure the safety and comfort of all program participants. This will be started with the "Heat and Smog" Inservice. All groups will be expected to modify Their activities to take account of the heat and are expected to spend at least one major activity a day in air-conditioned settings. This lasts typically two hours. They will also be expected to cool down their cars before loading passengers into them. They will offer water (not soda) frequently to consumers, provide hats and sunscreen, and know the symptoms of heat exhaustion and heat stroke and how to respond. During routine monthly inspections held by HDS administrators (the Program Service Review) the vehicles will be inspected for the presence of working air-conditioning, adequate cool water (at least one quart per person per day), window shades, seat covers when necessary. Staff will be quizzed on their preparedness to deal with heat related issues.